Sobolev Inequalities

<u>Q:</u> Are there exponents $q \in [1, \infty)$ st for all $u \in W^{1,p}(\mathbb{R}^n)$, $u \in L^q(\mathbb{R}^n)$ and

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)}$$

for some constant $C > 0$?

Suppose there is such a q. Then

$$\left(\int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \left(\int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p}.$$

Consider the scaling

$$u_\lambda(x) = u(\lambda x), \quad \lambda > 0.$$

We have

$$\int_{\mathbb{R}^n} |Du_\lambda(x)|^p \, dx = \int_{\mathbb{R}^n} \lambda^p |Du(\lambda x)| \, dx = \int_{\mathbb{R}^n} \lambda^{p-n} |Du(y)|^p \, dy$$

$$y = \lambda x$$

$$dy = \lambda^n dx$$

$$= \lambda^{p-n} \int_{\mathbb{R}^n} |Du|^p \, dx$$

$$u_\lambda(x) = u(\lambda x)$$

$$Du_\lambda(x) = \lambda Du(\lambda x)$$

$$\int_{\mathbb{R}^n} |u_\lambda|^q \, dx = \int_{\mathbb{R}^n} |u(\lambda x)|^q \, dx = \lambda^{-n} \int_{\mathbb{R}^n} |u|^q \, dx$$

So

$$\left(\int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \left(\int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p}$$

gives

$$\left(\lambda^{-n} \int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \left(\lambda^{p-n} \int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p}$$

$$\left(\int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \lambda^{\overset{\alpha}{\frac{n}{q}+1-\frac{n}{p}}} \left(\int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p} \qquad \forall \lambda > 0$$

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$A, B > 0$

$A \leq \lambda^{\alpha} B \quad \forall \lambda > 0 \Rightarrow \alpha$

If $\alpha > 0$, letting $\lambda \to 0$ gives $A \leq 0$, so $\alpha \leq 0$. If $\alpha < 0$, letting $\lambda \to \infty$ gives $A \leq 0$, so $\alpha = 0$.

So $\frac{n}{q} + 1 - \frac{n}{p} = 0$ and hence

$$q = \frac{np}{n-p}.$$

For $1 \leq p < n$,

q is called the <u>critical Sobolev exponent</u> and is denoted by p^*:

$$\boxed{p^* = \frac{np}{n-p}} \quad \left(\geq \frac{np}{n} = p \geq 1 \right)$$

Thm (Gagliardo-Nirenberg-Sobolev inequality):

If $1 \leq p < n$, then $\exists$ a constant $C > 0$ st

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)} \quad \forall u \in C_c^1(\mathbb{R}^n).$$

<u>Pf</u>: First we consider the case $p=1$. For simplicity, we take $n=2$. So we want to prove

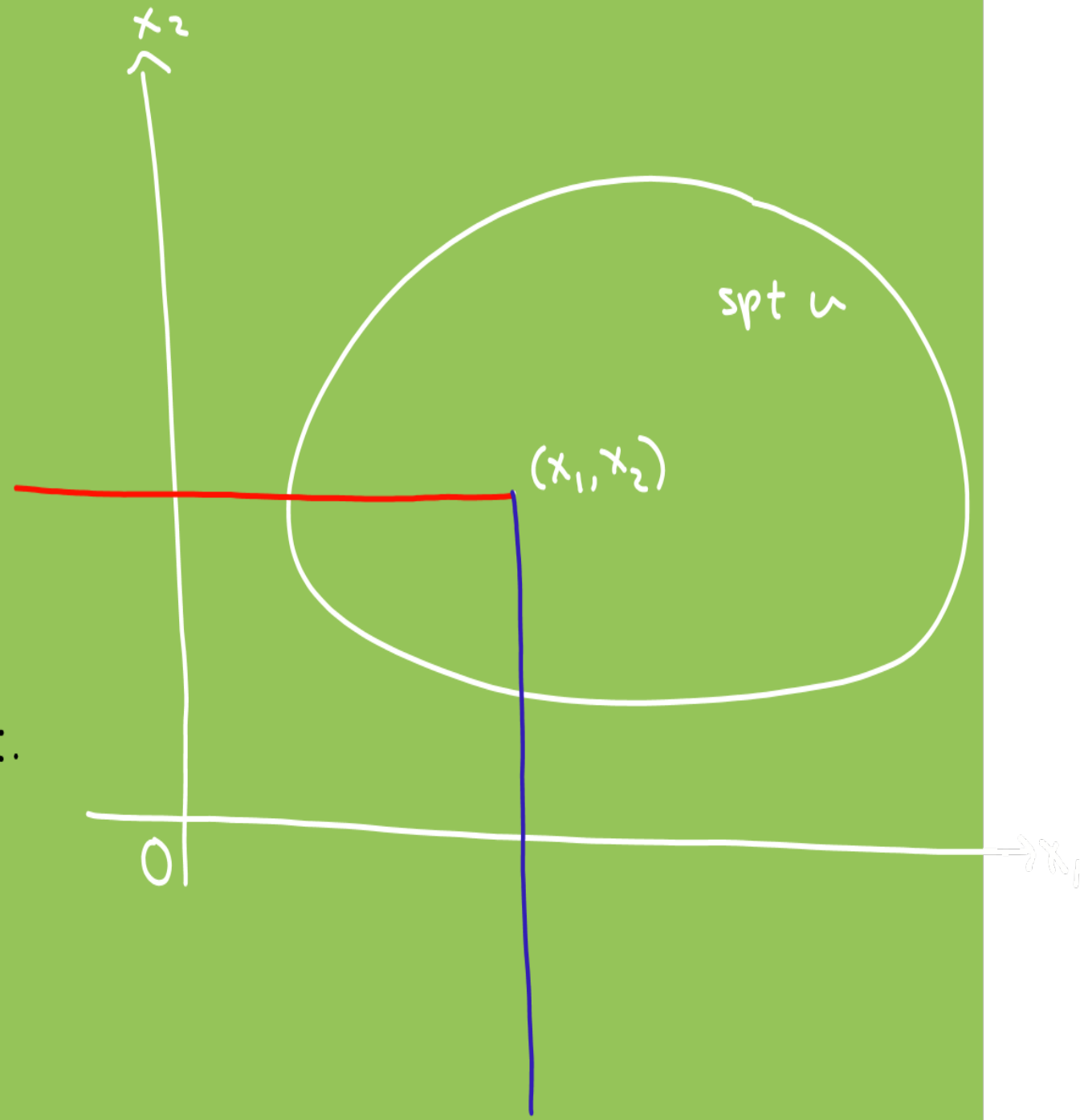
$$\left(\int_{\mathbb{R}^2} |u|^2 dx\right)^{1/2} \leq C \int_{\mathbb{R}^2} |Du| dx.$$

$$\boxed{p^* = \frac{np}{n-p} = 2 \quad \text{when } p=1 \text{ and } n=2}$$

$$u(x_1, x_2) = \int_{-\infty}^{x_1} u_{x_1}(y_1, x_2)\, dy_1$$

$$u(x_1, x_2) = \int_{-\infty}^{x_2} u_{x_2}(x_1, y_2)\, dy_2$$

since u has compact support.



$$|u(x_1,x_2)| = \left| \int_{-\infty}^{x_1} u_{x_1}(y_1,x_2)\,dy_1 \right|$$

$$\leq \int_{-\infty}^{x_1} |u_{x_1}(y_1,x_2)|\,dy_1$$

$$\leq \int_{-\infty}^{\infty} |Du(y_1,x_2)|\,dy_1$$

$$Du = (u_{x_1}, u_{x_2})$$

$$|Du| = \sqrt{u_{x_1}^2 + u_{x_2}^2}$$

$$\geq |u_{x_1}|$$

Similarly,

$$|u(x_1,x_2)| \leq \int_{-\infty}^{\infty} |Du(x_1,y_2)|\,dy_2.$$

Multiplying gives

$$|u|^2 \leq \left(\int_{-\infty}^{\infty} |Du(y_1, x_2)|\, dy_1\right)\left(\int_{-\infty}^{\infty} |Du(x_1, y_2)|\, dy_2\right).$$

So

$$\int_{\mathbb{R}^2} |u|^2\, dx \leq \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} |Du(y_1, x_2)|\, dy_1\right)\left(\int_{-\infty}^{\infty} |Du(x_1, y_2)|\, dy_2\right) dx_1\, dx_2$$

x_2 x_1

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du(y_1, x_2)| \, dy_1 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du(x_1, y_2)| \, dy_2 \, dx_1 \, dx_2$$

$$\int_{\mathbb{R}^2} |Du| \, dx$$

$$= \left(\int_{\mathbb{R}^2} |Du| \, dx \right) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du(y_1, x_2) \, dy_1 \, dx_2 = \left(\int_{\mathbb{R}^2} |Du| \, dx \right)^2$$